Exploring IT-driven supply chain capabilities and resilience: the roles of supply chain risk management and complexity

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Abstract

Purpose – In an era where global supply chains are increasingly susceptible to disruptions, this study aims to unlock the potential of IT integration within the supply chain in enhancing supply chain resilience. It investigates how this integration, combined with IT-driven capabilities, acts as a cornerstone in strengthening supply chain resilience (SCRs). The research navigates through the intricate dynamics of supply chain management to chart a strategic pathway for managers, illuminating the investments necessary to cultivate a resilient supply chain.

Design/methodology/approach – Grounding their investigation within the theoretical lens of dynamic resource-based theory, the authors designed a conceptual model to investigate the relationship between IT integration and SCR. Through the survey responses from supply chain professionals, the authors applied partial least squares-based structural equation modelling (PLS-SEM) to unravel the complexities of building a resilient supply chain.

Findings – Their investigation reveals that IT-driven supply chain capabilities (SCCs) are not just beneficial but essential in bridging IT integration with SCR. These capabilities emerge as vital, fully mediating the relationship between IT integration and SCR. Moreover, the significance of robust supply chain risk management (SCRM) in harnessing these IT-driven capabilities to bolster resilience is undeniable, offering organisations a sustainable advantage during times of turbulence.

Research limitations/implications – The primary limitation of this study is that data collection occurred during the COVID-19 pandemic. Therefore, the results may not be representative of different circumstances or periods. This study uncovered several noteworthy findings. The authors found that SCCs fully mediate the relationship between Supply Chain Management IT Integration (SCMITI) and SCR. In addition, the authors found that SCRM positively moderates the relationship between SCMITI and SCCs. The authors observed that Supply Chain Structural Complexity (SCSC) has no significant moderating effect on the relationship between IT-driven SCCs and SCR.

Practical implications – The results of this study offer managers a clear path forward for investing in SCR. Businesses must invest in SCMITI and SCRM to boost their SCR through developing IT-driven SCCs to ensure consistent business continuity. SCMITI does not necessarily increase SCR resilience when it is done poorly and does not lead to SCCs effectively. In this regard, managers must rely on collective wisdom to determine which technologies and IT applications will best meet their business needs and their peers' requirements in the supply chain.

Originality/value – This study sheds light on the mechanisms through which IT enhances SCR. It fills a critical knowledge gap, focusing on the synergy between IT integration, risk management and the cultivation of IT-driven capabilities to navigate supply chain disruptions.

Keywords Supply chain disruptions, Resilience, Risk management, Empirical study, Global supply chain

Paper type Research paper

1. Introduction

Since the post-World War II era, globalisation has incentivised businesses (Vanham, 2019) to offshore their manufacturing activities (Liu *et al.*, 2018). This shift has led to the formation of global supply chains and provided substantial profitability advantages for businesses. However, it has also increased the complexity and uncertainty of supply chains, making them more susceptible to disruptions – unplanned events that interrupt the flow of materials and goods (Chowdhury and Quaddus, 2016; Revilla and Saenz, 2017).

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The emergence of supply chain disruption risks has led to extensive exploration of the conventional risk management playbook, which encompasses risk identification, risk assessment and implementation of mitigation strategies (Braunscheidel and Suresh, 2009; Tummala and Schoenherr, 2011; Sodhi *et al.*, 2012). However, this traditional approach has proven to be inadequately scalable for addressing the complexities and dynamics within (global) supply chains (Wieland and Durach, 2021), as evidenced by the impact of recent disruptions.

For example, the COVID-19 pandemic in 2020, has significantly impacted the global economy (Herold *et al.*, 2021), shrinking the EU's real GDP by 6.1% (Verwey and Monk, 2021) and disrupting supply chains for 94% of Fortune 1000 companies (Sherman, 2020). Moreover, a McKinsey survey found that only 40% of corporate directors are adequately prepared for future disruptions (McLaughlin, 2022), and only 2% of companies have visibility beyond their second-tier suppliers (Alicke *et al.*, 2021).

In addition, the rise of the BRICS [1] nations towards a multipolar world order is expected to increase geopolitical instability and tensions, potentially leading to further geopolitical risks for global supply chains (Reivan-Ortiz *et al.*, 2023). Faced with a future marked by economic turbulence and the limitations of the conventional risk management playbook, supply chain partners must strengthen their Supply Chain Resilience (SCR) to manage future disruptions effectively.

Resilience enables supply chain partners to adapt to and overcome disruptions (Wieland and Durach, 2021). In this context, IT resources play a crucial role in creating higher-level dynamic IT capabilities. These dynamic IT capabilities drive superior performance over time (Winter, 2003) and are particularly needed in today's dynamic and turbulent business environment (Qaiyum and Wang, 2018).

Building on this understanding, SCM IT Integration (SCMITI) catalyses the creation of IT-driven supply chain capabilities (SCCs) by integrating IT platforms that store and transform various forms of information across the supply chain. This integration fosters resilience by facilitating the development of critical IT-driven SCCs, including flexibility (Chowdhury and Quaddus, 2017), visibility (Jüttner and Maklan, 2011; Mandal *et al.*, 2016), responsiveness (Braunscheidel and Suresh, 2009), collaboration (Mandal *et al.*, 2016) and integration (Rai *et al.*, 2006; Huo, 2012). These SCCs, in turn, significantly contribute to the enhancement of SCR, thereby enhancing the supply chain's ability to recover from disruptions.

Despite significant growth in the literature on SCR, research on how firms develop resilience to supply chain disruptions remains insufficient (Blackhurst *et al.*, 2011; Ambulkar *et al.*, 2015). In particular, there is limited knowledge about the underlying mechanisms by which IT enhances SCR. As Sodhi *et al.* (2012) emphasise, addressing these gaps through empirical research will enhance practitioners' understanding (van Hoek, 2020). This, in turn, enables managers to invest in appropriate IT tools and capabilities strategically, safeguarding business continuity and profitability (Sawyerr and Harrison, 2020).

This study integrates the resource-based view (RBV) and dynamic capability view (DCV) as dual complementary theoretical lenses (Wang *et al.*, 2024; Xia *et al.*, 2024) to conceptualise the constructs. We investigate the interplay mechanism between resources (i.e. IT resources), dynamic capabilities (i.e. IT-driven SCCs) and competitive advantage (i.e. SCR), as outlined by Huang *et al.* (2023). Therefore, we formulate the following Research Question (RQ):

RQ1. What is the role of SCMITI and IT-driven SCCs in developing SCR?

Besides the advantages that IT can provide in the form of dynamic IT capabilities, it also introduces vulnerabilities and risks such as cybersecurity threats, system failures and data breaches that can harm business continuity. Lloyd's of London, a central hub in cyber insurance, warns that a major cyberattack on a global payment system could cause $3.5tn in losses (Smith, 2023). To mitigate these risks, supply chain risk management (SCRM) must adopt robust IT risk management strategies and enhance cybersecurity (Cheung *et al.*, 2021).

Additionally, the attainment of benefits through IT may be hindered by supply chain structural complexity (SCSC), which can present challenges in monitoring and managing interactions among supply chain partners (Craighead *et al.*, 2007). Given the inconclusiveness of the empirical evidence on the role of SCSC in SCR (Akın Ateş *et al.*, 2022) and acknowledging the existing knowledge gap regarding the relationships between resilience, risk and SCM (Blackhurst *et al.*, 2005; Ponomarov and Holcomb, 2009), we formulate RQ2 as follows:

RQ2. How do SCRM and SCSC influence the development of SCR through IT-driven SCCs?

Our study makes substantial theoretical contributions by integrating RBV and DCV perspectives to elucidate how SCMITI enhances SCR, focusing on IT-driven SCCs, SCRM and SCSC. Practically, this study provides valuable insights for supply chain managers. It emphasises the importance of strategically investing in SCMITI to enhance SCR, highlighting that mere technology acquisition is insufficient. Managers must leverage IT investments to create unique capabilities that enhance resilience. The study guides IT platform selection to boost key SCCs and underscores the need for robust risk management practices, including cybersecurity measures. It advises a holistic approach to SCM, considering the dual impact of structural complexity on resilience.

The remainder of this paper is organised as follows: Section 2 covers the theoretical background. Section 3 presents the conceptual model and hypotheses formulation. The research methodology is described in Section 4. Section 5 reports data analysis and results, followed by discussions of implications and future research in Section 6. Finally, the paper is concluded in Section 7.

2. Theoretical background

In this section, we outline each higher-order construct of SCMITI, SCCs and SCR, drawing upon existing literature to

establish and define the necessary dimensions for their evaluation.

2.1 Supply chain management IT integration

Supply chains are conceptualised as dynamic networks, with nodes representing supply chain partners and links denoting their business interactions (Christopher, 2000; Lambert *et al.*, 2005; Craighead *et al.*, 2007; Pettit *et al.*, 2010) aimed at fulfilling customer requirements. In this context, SCMITI plays a critical role in facilitating the interaction among supply chain partners across the network to improve the flow of products, services, information, payments and decisions (Zhao *et al.*, 2008).

A focal supply chain partner with significant influence and resources often leads to initiating a common IT platform by leveraging its position to demonstrate the significant benefits that effective IT integration provides to the whole supply chain. This effort, combined with strategic alignment and collaborative efforts among supply chain partners, can significantly enhance the adoption and effectiveness of SCMITI (Rai *et al.*, 2006).

SCMITI refers to the extent to which a focal supply chain partner shares information with its upstream suppliers and downstream customers (Leuschner *et al.*, 2013; Wei *et al.*, 2020). In line with Rai *et al.* (2006), we conceptualise SCMITI as a second-order construct that encompasses two fundamental dimensions: Data Consistency (DC) and Cross-Functional Applications Integration (CFAI). DC refers to the extent to which common data definitions and consistency in stored structural data (e.g. inventory levels, demand forecasts, production schedules) have been established across a focal firm's supply chain. CFAI refers to the extent to which a partner communicates in real-time with other partners within the chain using SCM applications.

2.2 IT-driven supply chain capabilities

Building on the concept of organisational capabilities by Chakravarti and Day (1991) as a complex blend of skills and collective knowledge for achieving strategic goals (Bharadwaj, 2000), SCCs are defined as the ability of a focal supply chain partner to recognise, employ and incorporate internal and external resources to manage supply chain activities (Wu *et al.*, 2006). These capabilities can be categorised into lower-level operational capabilities and higher-level dynamic capabilities (Winter, 2003; Brusset and Teller, 2017).

Operational capabilities contribute to day-to-day operations, while dynamic capabilities enhance superior performance over time (Winter, 2003). Considering today's dynamic and turbulent business environment that favours dynamic capabilities (Qaiyum and Wang, 2018), this study concentrates exclusively on dynamic SCCs, specifically those driven by IT that contribute to SCR.

Wu *et al.* (2006) argue that integrated IT systems allow for real-time data sharing across departments, leading to improved decision-making and operational efficiency. Moreover, IT systems allow for advanced analytics and tools to rapidly adjust supply chain processes in response to sudden market changes (Wu *et al.*, 2006). Additionally, IT systems enhance the ability to sense market changes and seise opportunities, a core aspect of dynamic capabilities (Wu *et al.*, 2006).

Therefore, the abovementioned examples support that IT systems facilitate data integration and operational adjustments, reflecting the adaptive nature of dynamic capabilities. Through the theoretical lens of the dynamic RBV, we conceptualise IT-driven SCCs in the SCCs-SCR relationship as a second-order construct that encompasses five dimensions: flexibility (Chowdhury and Quaddus, 2017), visibility (Mandal *et al.*, 2016), responsiveness (Wu *et al.*, 2006; Yu *et al.*, 2018), collaboration (Mandal *et al.*, 2016) and integration (Wu *et al.*, 2006; Yu *et al.*, 2018). These five SCCs stem from the strategic utilisation and integration of IT. We briefly characterise each dimension below.

Supply Chain Flexibility (SCFL) is defined as the ability of a focal supply chain partner to swiftly and efficiently deploy diverse resources to address market changes (Jüttner and Maklan, 2011; Dubey *et al.*, 2021). The resources encompass a diverse array, including financial assets, human capital, technological tools and physical infrastructure. SCFL enables supply chain partners to encounter, resolve and exploit unexpected emergencies in supply, manufacturing and distribution processes (Sreedevi and Saranga, 2017).

Supply Chain Visibility (SCVI) is described as the ability of a focal supply chain partner to have access to or share timely information about its operations (Jüttner and Maklan, 2011; Dubey *et al.*, 2021). It involves the timely exchange of information about entities moving through the supply chain and the planned and actual times of events (Jüttner and Maklan, 2011). This information encompasses real-time inventory levels, demand forecasts, production schedules, shipment statuses and any disruptions or delays, among others. SCVI uncovers the location of resources within the chain and communicates this information to SC partners (Blackhurst *et al.*, 2011).

Supply Chain Responsiveness (SCRS) is defined as the ability of a focal supply chain partner to adapt to customer requirements and cope with variations in business conditions and market fluctuations (Klibi *et al.*, 2010; Sabouhi *et al.*, 2020). Using SCRS, supply chain partners can offer product variety to their downstream customers in a time-effective manner (adapted from Holweg (2005) and Reichhart and Holweg (2007)).

Supply Chain Collaboration (SCCL) is defined as the ability of a focal supply chain partner to make decisions jointly and work with its partners in the chain (Jüttner and Maklan, 2011). SCCL significantly enhances information-processing capacity (Dubey *et al.*, 2021) by enabling supply chain partners to share information, and jointly plan, coordinate and execute activities. This collaboration aims to achieve a seamless flow of goods, information and finances across the supply chain (Zhao *et al.*, 2008).

Supply Chain Integration (SCIN) is defined as a supply chain partner's ability to integrate its internal and external organisational activities with those of its supply chain partners (Huo, 2012). This integration is crucial for effectively synchronising, coordinating and optimising the main flows within the supply chain (Zhao *et al.*, 2008).

2.3 Supply chain resilience

The multidimensional and multidisciplinary concept of resilience in SCM is relatively new, having been introduced in

the early 2000s (Starr *et al.*, 2003; Christopher and Peck, 2004; Christopher and Rutherford, 2004; Sheffi and Rice, 2005). SCR refers to the supply chain's ability to effectively manage disruption risks and recover from them, returning to the original or a better state (Christopher and Peck, 2004; Chowdhury *et al.*, 2019). In this context, resilience enables supply chain partners to adapt in response to disruptions (Wieland and Durach, 2021). This requires both readiness and the ability to respond and recover, to prevent significant negative impacts on the supply chain.

Robust supply chain processes that can absorb disruptions can be developed through resilience, rather than attempting to resist vulnerabilities (Pettit *et al.*, 2010). This can be done by aligning resources to adapt to the business environment strategically and outperforming competitors (Hamel and Välikangas, 2003). Drawing on stability- and adaptation-based perspectives of SCR as highlighted by Ivanov (2023), we conceptualise SCR as a second-order construct reflected in three dimensions: recovery (Chowdhury and Quaddus, 2017), agility (Liu *et al.*, 2018) and environmental sustainability (Marchese *et al.*, 2018). Recovery and agility, rooted in the stability-based view, focus on restoring normalcy post-disruption. However, environmental sustainability, reflecting the adaptation perspective, enhances long-term supply chain viability through innovative practices and responsible resource management.

Supply Chain Recovery (SCRE) refers to the ability of a focal supply chain partner to return to a normal level of functionality after a disruption and is widely recognised as a key component of SCR (Sheffi and Rice, 2005). SCRE's pivotal role in maintaining operational stability and mitigating disruption consequences solidifies its position as a key metric for measuring overall SCR.

Supply Chain Agility (SCAG) refers to the ability of supply chain partners to respond to changes in the external environment effectively (Liu *et al.*, 2018). Agility triggers unintentional learning across supply chain partners by facilitating the transfer of unconventional knowledge that may not have been formally shared among them before the disruption (Scholten *et al.*, 2019). This knowledge transformation allows for better coordination and quicker responses to disruptions, thereby enhancing the overall SCR.

Supply Chain Sustainability (SCSU) is described as the ability of a focal supply chain partner to mitigate negative environmental impacts and address sustainability issues, integrating SCM with environmental management (Fiksel, 2006; Kusi-Sarpong *et al.*, 2019; Rajesh, 2021). Marchese *et al.* (2018) argue that a more sustainable system inherently becomes more resilient. Therefore, SCSU's ability to absorb and mitigate disruptive environmental challenges positions it as a crucial indicator of SCR.

3. Conceptual model and hypotheses development

Despite significant growth in the SCR literature, critical gaps persist in understanding how firms develop resilience (Blackhurst *et al.*, 2011; Ambulkar *et al.*, 2015) and particularly the role of IT resources in enhancing SCR. As Sodhi *et al.* (2012) highlighted, more empirical research is needed to address these gaps. Moreover, existing studies lack conclusive empirical evidence on the role of SCSC in SCR (Akın Ateş *et al.*, 2022) and exhibit a knowledge gap regarding the relationships between resilience, risk and SCM (Blackhurst *et al.*, 2005; Ponomarov and Holcomb, 2009).

To bridge these gaps, similar to Wang *et al.* (2024) and Xia *et al.* (2024), this study integrates the RBV (Wernerfelt, 1984; Barney, 1991) and DCV (Teece *et al.*, 1997) as dual complementary theoretical lenses. Within the RBV framework, we identify lower-level tangible IT resources (e.g. technology and IT infrastructure), alongside complementary intangible resources (e.g. human knowledge and skills in IT), that collectively contribute to the development of IT-driven capabilities (Bharadwaj, 2000; Rai *et al.*, 2006).

We then extend this interpretation through the lens of DCV by conceptualising IT-driven SCCs as dynamic capabilities, enhancing SCR and thereby serving as a crucial competitive advantage. In other words, SCR enables the supply chain partners to effectively manage disruptions and meet market demands more efficiently than competitors during times of turbulence (Dubey *et al.*, 2021). This application of the dynamic RBV, aligned with the DCV principles, aims to offer a clearer understanding of the interplay between resources (i.e. SCMITI), dynamic capabilities (i.e. IT-driven SCCs) and competitive advantage (i.e. SCR).

Consistent with the framework on the relationship between IT, capabilities and competitive advantage outlined by Huang *et al.* (2023), Figure 1 illustrates our hypotheses by including higher-order constructs of SCMITI, IT-driven SCCs and SCR as well as SCRM and SCSC as two moderators.

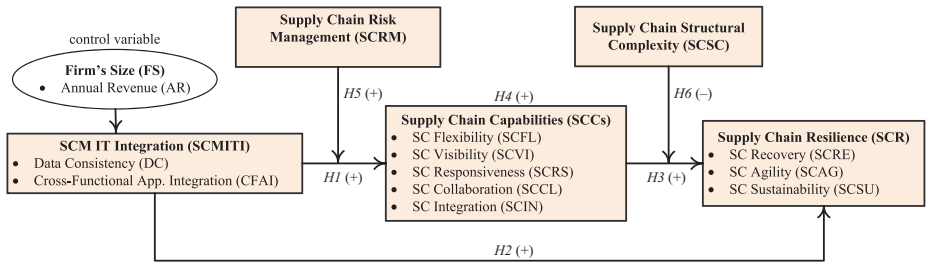
3.1 The effect of supply chain management IT integration

The RBV framework posits that firms gain a competitive advantage through unique, valuable and hard-to-imitate resources (Wernerfelt, 1984; Barney, 1991). SCMITI embodies such resources by integrating lower-level tangible IT assets, such as technology and IT infrastructure, with complementary intangible resources, such as human knowledge and IT skills (Bharadwaj, 2000; Rai *et al.*, 2006). Drawing upon RBV theory, we argue that SCMITI leverages these collective resources to foster the development of IT-driven capabilities. According to Bharadwaj (2000) and Wu *et al.* (2006), these IT-driven capabilities provide a competitive edge when their replication by competitors becomes more challenging. Therefore, within the RBV framework, we propose the following hypothesis:

H1. SCMITI has a positive impact on the development of SCCs.

Moreover, the DCV (Teece *et al.*, 1997) extends RBV by emphasising a firm's ability to integrate internal and external competencies and resources to address rapidly changing environments. In this context, SCMITI provides opportunities for blurring the boundaries between supply chain partners, enabling them to reduce uncertainty by aligning their inter-organisational processes with one another (Vanpoucke *et al.*, 2017). As an enabler of information exchange, IT plays a major role in supply chain activities integration by allowing supply

Figure 1 The conceptual model and hypotheses



Source: Authors' own work

chain partners to increase the volume and complexity of information exchange and share real-time information (Vanpoucke *et al.*, 2017). By sharing and exchanging information on various cross-functional and cross-firm business activities (e.g. demand forecasts, inventory levels and customer needs), SCMITI helps supply chain partners collaborate and respond to environmental changes (Wei *et al.*, 2020). Therefore, within the DCV framework, we propose the following hypothesis:

H2. SCMITI has a positive impact on the development of SCR.

3.2 The effect of IT-driven supply chain capabilities

Drawing upon the DCV (Teece *et al.*, 1997), which emphasises the integration of dynamic capabilities to create competitive advantage, we posit that IT-driven SCCs significantly enhance firms' competitive advantage during turbulent times by bolstering SCR. These capabilities include flexibility (Chowdhury and Quaddus, 2017), visibility (Mandal *et al.*, 2016), responsiveness (Wu *et al.*, 2006; Yu *et al.*, 2018), collaboration (Mandal *et al.*, 2016) and integration (Wu *et al.*, 2006; Yu *et al.*, 2018). Each of these IT-driven SCCs contributes to supply chain partners' ability to adapt, respond and thrive in dynamic environments, thereby fostering resilience and sustaining competitive advantage.

Specially, IT-driven capabilities provide supply chain partners with real-time tracking of demand and product availability across the supply chain, and in turn, empower faster responses to shifts in demand (van Hoek, 2020). Wu *et al.* (2006) argue that IT systems facilitate data integration and operational adjustments, reflecting the adaptive nature of dynamic capabilities (Wu *et al.*, 2006). Moreover, these capabilities are crucial for active learning about customer needs, competitor strategies and market trends, enabling supply chain partners to adapt their operations in response (Aslam *et al.*, 2018). Therefore, we argue that IT-driven SCCs contribute significantly to SCR by developing and maintaining competitive advantage in dynamic market conditions. Based on this argumentation, we formulate the following hypothesis:

H3. SCCs have a positive impact on the development of SCR.

By combining the aforementioned arguments and the one presented in Subsection 3.1, we posit a comprehensive theoretical framework that integrates the RBV and the DCV to explain the relationship between SCMITI and SCR through the development of IT-driven SCCs. Initially, we established that SCMITI contributes to the development of SCCs within the RBV framework. Subsequently, we argued that SCCs, as dynamic capabilities under the DCV, play a crucial role in developing competitive advantage in the form of enhanced SCR. These arguments align with Irfan *et al.* (2020), who argue that IT integration enables supply chain flexibility (i.e. a dimension of SCCs), which in turn results in higher supply chain agility (i.e. a dimension of SCR). Therefore, we develop the following hypothesis:

H4. SCCs positively mediate the relationship between SCMITI and SCR.

3.3 The effect of supply chain risk management

Modern supply chains are ranked among the most vulnerable systems (Guerra *et al.*, 2024), necessitating robust SCRM. While the literature on SCRM has been extensively explored, there is no clear consensus on its definition (Sodhi *et al.*, 2012). For this study, we define SCRM as the process of identifying risk factors, assessing them and implementing mitigation strategies (Braunscheidel and Suresh, 2009; Tummala and Schoenherr, 2011; Sodhi *et al.*, 2012). As the supply chains rapidly evolve through the integration of IT, the traditional SCRM needs a paradigm shift (Colicchia *et al.*, 2019; Farrell and Gallagher, 2019) and a more holistic and integrated view of the SCRM is required (Heckmann *et al.*, 2015; Guerra *et al.*, 2024).

The interconnectivity of organisational processes within and across supply chain partners has the potential to foster IT-driven SCCs (Vanpoucke *et al.*, 2017). However, it could also introduce new types of supply chain vulnerabilities, such as cyber and informational risks (Creazza *et al.*, 2022). Therefore, SCRM plays a crucial role in the effectiveness of leading SCMITI to IT-driven SCCs by mitigating the risks associated with IT. By providing a robust SCRM framework organisations can proactively manage and mitigate the risks (Ambulkar *et al.*, 2015) that can potentially undermine the effectiveness of SCMITI in creating SCCs. Therefore, following the key role of SCRM in protecting and strengthening the relationship between SCMITI and SCCs, we formulate the following hypothesis:

H5. SCRM strengthens the relationship between SCMITI and SCCs.

3.4 The effect of supply chain structural complexity

Modern supply chains are ranked as some of the most complex in the contemporary world (Guerra *et al.*, 2024). Complexity in systems can lead to increased uncertainty, which consequently reduces clarity and control, making it difficult for business leaders to navigate and manage their organisations effectively (Reeves *et al.*, 2020). As businesses diversify their products, adopt new technologies and extend their worldwide supply networks, supply chains will inevitably take more complex structures (Akın Ateş *et al.*, 2022). Structural complexity is related to uncertainty originating from static characteristics of the supply chain such as the degree of connections among partners and the diversity of partners (Cheng *et al.*, 2014; Bode and Wagner, 2015). In line with Chowdhury *et al.* (2019) and Craighead *et al.* (2007), we define SCSC as a measure of the interconnectedness of a supply chain partner with their upstream suppliers and downstream customers.

According to Akın Ateş *et al.* (2022), the impact of SCSC on business performance remains uncertain. Some studies have discovered a negative impact (Brandon-Jones *et al.*, 2015), while others have observed a positive impact (Sharma *et al.*, 2019) or no impact at all (Chaudhuri and Boer, 2016). Reeves *et al.* (2020) argue that SCSC may hinder efficiency but it can improve resilience and adaptability by increasing the diversity of the business partners. It has been observed that complexity can amplify disruptions (Craighead *et al.*, 2007), worsen operational performance, complicate decision-making and lead to disruptions (Bode and Wagner, 2015). Nonetheless, the empirical evidence concerning the impact of SCSC on SCR remains unresolved. Organisations can leverage IT-driven SCCs to enhance the effectiveness of risk-mitigating measures (e.g. multiple sourcing, backup plans, etc.); however, uncertainty can reduce the effectiveness of these measures. Thus, SCSC can weaken the relationship between SCCs and SCR. We propose the following hypothesis:

H6. SCSC weakens the relationship between SCCs and SCR.

3.5 Control variable

Consistent with Mabert *et al.* (2003) and Revilla and Saenz (2017), we incorporated firm size (FS) as it may influence the main variables in the conceptual model and introduce unwanted sources of variance.

4. Research methodology

4.1 Sample and data collection

In this study, we employed the questionnaire-based survey method to adopt items from the existing literature (Dennehy *et al.*, 2021). We measured the constructs using survey items adapted from the literature. The unit of analysis was a random selection of professionals from all tiers of the supply chain, chosen to provide a comprehensive understanding of the supply chain and its performance, as demonstrated in previous research (Brandon-Jones *et al.*, 2014).

The survey was designed using Qualtrics which is a web-based platform (Dennehy *et al.*, 2021). Before distribution, the survey was reviewed by three faculty members and pretested then made available online from 18 August 2020, to 18 December 2020. The targeted sample of respondents was composed of supply chain professionals from around the world who were contacted via LinkedIn. A total of 412 respondents completed the survey anonymously, but 18 of them did not fill out the survey completely and were therefore removed due to person-level missingness. The demographic profile of the remaining respondents is provided in Table 1.

4.2 Measurement development

All constructs and survey items were measured using a five-point Likert scale, a common practice in empirical research (Qin *et al.*, 2021). To enhance scale performance, minor modifications were made to some items (Dennehy *et al.*, 2021). Details of the survey items and their literature sources are provided in Table A1 in the Appendix.

Independent variable. In line with Rai *et al.* (2006), we modelled the latent construct SCMITI as a second-order construct consisting of the sub-constructs DC and CFAI. We measured DC using a combination of items adapted from van den Hoven (2004), Rai *et al.* (2006), Lu and Ramamurthy

Table 1 Demographic profile of the respondents ($n = 394$)

Title	Frequency	%
1. *Education level*		
Bachelor's degree	101	25.63
Master's degree	250	63.45
PhD	23	5.84
Others	20	5.08
2. *Job level*		
Junior	126	31.98
Manager	130	32.99
Senior manager	101	25.63
Top manager	37	9.39
3. *Years of experience*		
Less than 5 years	136	34.52
Between 5 and 10 years	82	20.81
Between 11 and 15 years	46	11.68
More than 15 years	130	32.99
4. *Location of the company*[a]		
Asia	102	25.89
Africa	9	2.28
America	79	20.05
Australia	2	0.51
Europe	201	51.02
5. *Annual revenue*		
Less than €2 Million	34	8.63
€2–9.9 Million	44	11.17
€10–49.9 Million	52	13.20
€50–99.9 Million	37	9.39
€100–499.9 Million	48	12.18
€500–999.9 Million	43	10.91
€1 Billion and more	136	34.52

Notes: [a]One respondent did not mention the location of their company. Thus, the total percentage is not 100%
Source: Authors' own work

(2011) and Guo *et al.* (2014). These items related to the use of automatic data capture mechanisms, data definition and storage consistency, the application portfolio and data management and architectures and compliance with international data content standards through the use of RFID technology. We measured the CFAI sub-construct by adapting items from Rai *et al.* (2006) and Yu *et al.* (2018), which related to the applications used for supply chain planning and transactions, as well as their interaction with internal applications and database consistency across functional departments.

Dependent variable. The second-order construct SCR and its three sub-constructs – SCAG, SCSU and SCRE – were modelled. SCAG was measured with five items adapted from Liu *et al.* (2018), reflecting a firm's rapid response to market changes and customer needs. SCSU was measured using six items adapted from Kusi-Sarpong *et al.* (2019), emphasising the environmental dimension of sustainability. The items reflect sustainable research and development, sustainable waste management, efficient energy consumption, eco-friendly product distribution, green product market responsiveness and socio-environmental standards implementation. SCRE was measured using four items adapted from Chowdhury and Quaddus (2017), reflecting the firm's capability to manage disruptions with minimal losses and quick, cost-effective recovery.

Mediator variable. The second-order construct, SCCs, was modelled in terms of its five sub-constructs: SCFL, SCVI, SCRS, SCCL and SCIN. SCFL was measured using seven items adapted from Chowdhury and Quaddus (2017), focusing on production, product diversity, workforce, sourcing and product development. SCVI was measured using three items adapted from Mandal *et al.* (2016) related to the firm's information accessibility for monitoring, as well as two items adapted from Jüttner and Maklan (2011) related to tracking systems. SCRS was measured with five items adapted from Braunscheidel and Suresh (2009) related to market-responsive order decisions. SCCL was measured using four items adapted from Mandal *et al.* (2016) related to strategic cooperation with suppliers. Finally, SCIN was measured using three items adapted from Huo (2012) related to real-time internal connectivity, as well as two items adapted from Rai *et al.* (2006) related to information sharing with partners.

Moderator variables. SCRM was measured using three items adapted from Ambulkar *et al.* (2015) and one item adapted from Sreedevi and Saranga (2017). These items collectively evaluate the firm's capability to manage risks and disruptions within the supply chain. Specifically, they measure the existence of a dedicated department or team for risk management, the utilisation of Key Performance Indicators (KPIs) and metrics for monitoring risks, the deployment of an information system tailored to manage disruptions and the proactive detection of operational risks through supplier monitoring, inspections and tracking. SCSC was measured using four items adapted from Chowdhury and Quaddus (2017). These items capture the complexity of the supply chain structure by examining the firm's approach to reducing complexity through direct engagement with suppliers and buyers, the limited bidirectional flow of goods and services, the use of multiple suppliers to mitigate supply risks and the strategy of engaging multiple buyers to minimise disruption risks. The items reflect the organization's efforts to manage and streamline the intricate dynamics associated with supplier and buyer relationships within the supply chain.

Control variable. In this study, we also included FS as a control variable, as it may influence the main variables in the conceptual model and introduce unwanted sources of variance. Consistent with Mabert *et al.* (2003) and Revilla and Saenz (2017), we measured FS using Annual Revenue (AR) as a proxy, as firms with more annual revenue tend to have a stronger financial position and may be able to invest more in IT, potentially leading to higher levels of SCMITI compared to firms with less annual revenue.

5. Data analysis and results

5.1 Analysis strategy and model evaluation

Partial least squares-based structural equation modelling (PLS-SEM) was used for the analyses using SmartPLS Version 4 (Ringle *et al.*, 2022), which is suited for complex models and estimating nomological networks among latent variables and is frequently used in operations management research (Hair *et al.*, 2011; Doering and Suresh, 2016; Akter *et al.*, 2021; Dennehy *et al.*, 2021). To validate the hierarchical component model, the disjoint two-stage approach was used (Becker *et al.*, 2012; Sarstedt *et al.*, 2019). A measurement model with lower-order components was constructed first in stage one. To achieve satisfactory reliability and validity, seven items were removed from some constructs (the italic rows in Table A1 in the Appendix): one item from the DC scale of SCMITI, three items from the SCFL scale of SCCs, one item from the SCRE scale of SCCs, one item from the SCSC scale and one item from the SCAG scale of SCR. After removing these items, the model characteristics yielded sufficient scores for the composite reliabilities of the constructs (greater than 0.70, ranging from 0.77–0.91, but the Cronbach's alpha was below the desired level) and sufficient convergent validity (Fornell and Larcker, 1981) with AVE's ranging from 0.51–0.71 (see Table 2).

As can be observed in Table 2, discriminant validity was established using the Fornell–Larcker criterion. The alternative HTMT ratios were investigated as well (not shown). All HTMT ratios were under the 0.9 threshold (Henseler *et al.*, 2015). No indication was found for concerns of common method variance in the data based on Harman's one-factor test (Podsakoff *et al.*, 2003). In addition, no correlation between constructs exceeded 0.90 (see Table 2) and no variance inflation factor (VIF) exceeded 3.3 (Bagozzi *et al.*, 1991). For stage two, scores of all lower-order variables were saved and added as new variables to the data set where the construct scores were used as indicators for the higher-order construct measurement model (Sarstedt *et al.*, 2019).

For the reflective higher-order constructs, sufficient scores were obtained for the composite reliabilities (> 0.70) and the AVEs ranged from 0.53–0.69 (see Table 3). For the formative higher-order construct, both indicators were significant ($p < 0.001$), and the VIF values were below 3.3 (Diamantopoulos and Siguaw, 2006). Discriminant validity was also established based on both the Fornell-Larcker criterion and the HTMT ratio (Fornell and Larcker, 1981; Henseler *et al.*, 2015).

Table 2 Composite reliability (CR) and correlations for stage one with the square root of average variance extracted (AVE) in parentheses

Construct	CR	AVE	1	2	3	4	5	6	7	8	9	10	11	12
1. SCAG	0.82	0.54	(0.74)											
2. SCRS	0.88	0.64	0.56	(0.80)										
3. CFAI	0.85	0.53	0.31	0.32	(0.73)									
4. SCCL	0.91	0.71	0.45	0.49	0.44	(0.85)								
5. DC	0.86	0.51	0.35	0.36	0.68	0.42	(0.72)							
6. SCFL	0.81	0.52	0.45	0.46	0.41	0.44	0.35	(0.72)						
7. SCIN	0.85	0.52	0.47	0.46	0.59	0.57	0.56	0.40	(0.73)					
8. SCRE	0.86	0.61	0.52	0.48	0.29	0.40	0.33	0.41	0.39	(0.78)				
9. SCSC	0.77	0.53	0.44	0.50	0.36	0.46	0.33	0.48	0.34	0.36	(0.73)			
10. SCRM	0.90	0.69	0.48	0.45	0.51	0.49	0.48	0.37	0.52	0.44	0.30	(0.83)		
11. SCSU	0.91	0.61	0.51	0.45	0.39	0.43	0.44	0.42	0.43	0.49	0.42	0.49	(0.78)	
12. SCVI	0.86	0.55	0.51	0.52	0.53	0.62	0.51	0.38	0.70	0.38	0.36	0.57	0.50	(0.74)

Source: Authors' own work

Table 3 Composite reliability (CR) and correlations for stage two with the square root of average variance extracted (AVE) in parentheses

Construct	CR	AVE	1	2	3	4	5
1. SCMITI	–	–	–				
2. SCSC	0.77	0.53	0.38	(0.73)			
3. SCRM	0.90	0.69	0.54	0.30	(0.83)		
4. SCCs	0.90	0.59	0.51	0.56	0.73	(0.77)	
5. SCR	0.86	0.67	0.46	0.50	0.58	0.73	(0.82)

Source: Authors' own work

5.2 Hypotheses testing

To assess the higher-order model, the path effects were estimated with bias-corrected bootstrapping and 5,000 subsamples. The Standardised Root Mean Square Residual for the estimated model was 0.072, indicating an acceptable fit (Cho *et al.*, 2020). All path effects are presented in Table 4.

We found SCMITI was a significant predictor of SCCs, $\gamma = 0.42$, $p < 0.001$, which supports *H1*. No significant direct effect was found between SCMITI and SCR, $\gamma = 0.03$, $p = 0.628$. Hence, no support was found for *H2*. A significant indirect effect was found between SCMITI and SCR, $\gamma = 0.25$, $p < 0.001$, implying full mediation. SCCs were found to be significantly related to SCR, $\gamma = 0.60$, $p < 0.001$, supporting H3. The SCRM was found to be a significant predictor of SCCs, $\gamma = 0.42$, $p < 0.001$.

The SCSC was a significant predictor of SCR with $\gamma = 0.17$ and $p < 0.001$. The SCCs are found to be a significant mediator. The indirect effect of SCMITI on SCR had a coefficient of 0.25 with $p < 0.001$, indicating full mediation, supporting H4. The interaction effect of SCRM and SCMITI (i.e. SCRM×SCMITI) on SCCs was significant with $\gamma = 0.08$, $p = 0.032$, supporting H5, as visualised in Figure 2, indicating that a higher level of SCRM strengthens the positive relationship between SCMITI and SCCs.

The proposed interaction effect between SCSC and SCCs (i.e. SCSC×SCCs) on SCR was not significant, $\gamma = 0.03$, $p = 0.555$. Hence, no support was found for H6. The adjusted R-squares were 0.52 and 0.51 for SCCs and SCR, respectively, with all Q_Predict^2 scores above zero (Shmueli *et al.*, 2019). Moreover, we found that the control variable FS in terms of annual sales has a significant impact on SCMITI, meaning that firms with higher annual sales can achieve a higher level of SCMITI. Figure 3 illustrates the path model of this study analysed using PLS-SEM.

6. Discussion, implications and future research

6.1 Theoretical implications

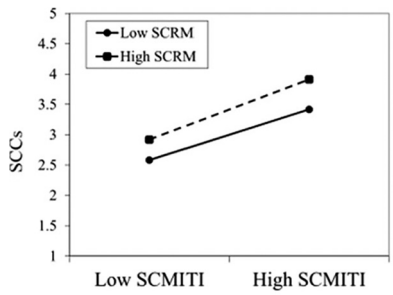
Our analysis yielded several significant findings. Primarily, we discovered that IT-driven SCCs fully mediate the relationship

Table 4 Path coefficients and predictive power f^2

Hypotheses	Relationships	Coefficient (γ)	*t*-statistics	*p*-values	f^2-values	Support
H1	SCMITI → SCCs	0.42	8.01	0.000	0.26	Yes
H2	SCMITI → SCR	0.03	0.44	0.663	0.00	No
H3	SCCs → SCR	0.60	12.35	0.000	0.37	Yes
	SCRM → SCCs	0.42	8.82	0.000	0.24	
H4	SCMITI → SCCs → SCR	0.25	6.86	0.000	–	Yes
H5	SCRM×SCMITI → SCCs	0.08	2.19	0.028	0.02	Yes
	SCSC → SCR	0.17	3.88	0.000	0.04	
H6	SCSC × SCCs → SCR	0.03	0.60	0.548	0.00	No
	Annual sales → SCMITI	0.33	6.75	0.000	0.12	

Source: Authors' own work

Figure 2 Moderating effect SCMITI × SCRM on SCCs



Source: Authors' own work

between SCMITI and SCR. This mediation confirms direct relationships between SCMITI and IT-driven SCCs, as well as between IT-driven SCCs and SCR while revealing no direct relationship between SCMITI and SCR.

The confirmed direct relationship between SCMITI and IT-driven SCCs (H1) aligns with existing literature, indicating that IT integration facilitates the development of SCCs by enhancing coordination, information sharing and efficiency. Wu *et al.* (2006) discovered that advancements in IT and the alignment of IT resources can facilitate the development of SCCs. Irfan *et al.* (2020) demonstrated that IT is a crucial resource for process integration that leads to flexibility capability. Similarly, Poberschnigg *et al.* (2020) found that cross-functional integration factors generate specific SCCs that influence resilience. Fawcett *et al.* (2011) noted that targeted IT investment promotes dynamic collaboration capabilities within the supply chain, thereby enhancing firm performance. Vanpoucke *et al.* (2017) further emphasised that IT is a crucial enabler for supply chain integration, allowing partners to increase the volume and complexity of real-time information sharing.

Interestingly, we found no direct relationship between SCMITI and SCR (H2). This absence aligns with the findings by Bharadwaj (2000) and Wu *et al.* (2006), who argued that IT investments do not inherently lead to competitive advantage unless the firm leverages their investment to create unique capabilities. This finding is also supported by Yu *et al.* (2018), who suggested that resources should be used in implementing strategies that are not easy to duplicate by competitors.

Moreover, the direct relationship between IT-driven SCCs and SCR (H3) is supported by literature that underscores the beneficial impact of IT-based decision-making on SCR. Belhadi *et al.* (2022) demonstrated that IT-enabled decision-making processes enhance resilience by allowing firms to respond more swiftly and effectively to disruptions. Nandi *et al.* (2020) identified information sharing, collaboration, integration and coordination as key capabilities of blockchain technology that can enhance service quality and flexibility.

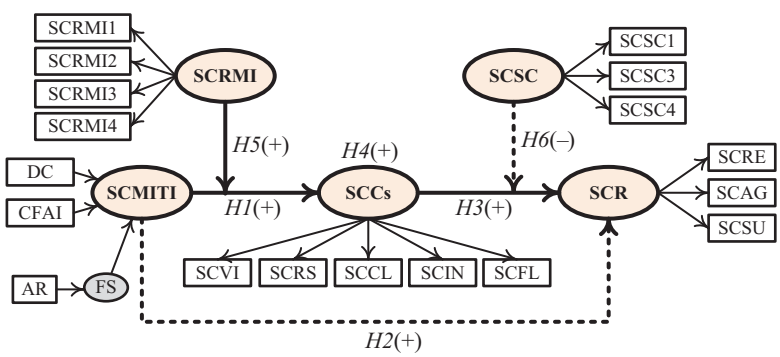
According to the discussion above, confirming the fully mediated relationship between SCMITI and SCR by IT-driven SCCs (H4) highlights the crucial role of these capabilities in enhancing SCR. IT-driven SCCs such as flexibility (Chowdhury and Quaddus, 2017), visibility (Jüttner and Maklan, 2011; Mandal *et al.*, 2016), responsiveness (Braunscheidel and Suresh, 2009), collaboration (Mandal *et al.*, 2016) and integration (Rai *et al.*, 2006; Huo, 2012) serve as essential links in translating IT resources and investments into IT-driven SCCs that ultimately lead to resilience outcomes.

Our analysis revealed a positive moderating effect of SCRM on the relationship between SCMITI and IT-driven SCCs (H5). This finding aligns with the proactive role of SCRM in managing and mitigating risks, as noted by Ambulkar *et al.* (2015). They argue that SCRM provides a robust framework for identifying, assessing and mitigating risks. Since the role of IT in SCM is inevitable, balancing IT integration with IT security should be considered to leverage IT capabilities effectively as emphasised by Smith *et al.* (2007).

In light of Wang-Mlynek and Foerstl's (2020) consideration that no single company is immune to risk in today's dynamic business landscape and the critical shift from an individual to a holistic view of risk management highlighted by Colicchia *et al.* (2019), we consider cybersecurity as a "must-have" within SCRM. This shift is particularly crucial given the escalation of supply chain cyber risks, evidenced by high-profile attacks (Creazza *et al.*, 2022) and hackers' increased sophistication in breaching IT defences, facilitated by generative AI. This finding underscores SCRM's role in enhancing IT-driven SCC development and, consequently, improving SCR.

Contrary to expectations, we found that supply chain structural complexity has no significant moderating effect on the relationship between IT-driven SCCs and SCR (H6). This finding aligns with previous research by Caniato and Größler (2015) and Chaudhuri and Boer (2016), which showed

Figure 3 The path model



Source: Authors' own work

SCSC's neutral impact on firm performance. Although the impact of SCSC on performance remains inconclusive (Akın Ateş *et al.*, 2022), it can be argued that the structural dimension of complexity in the supply chain can both weaken and strengthen the relationship simultaneously.

On one hand, as Reeves *et al.* (2020) argued, structural complexity can enhance resilience by providing more opportunities for adaptability within the ecosystem and increasing resilience. On the other hand, structural complexity increases the frequency of disruptions (Bode and Wagner, 2015), which can lead to less SCR. Therefore, the positive and negative impacts of SCSC may cancel each other out, resulting in no significant moderating effect on the relationship between SCCs and SCR.

6.2 Practical implications

This study provides practical insights to supply chain managers, emphasising the critical importance of investing in SCMITI to enhance SCR. Particularly, it highlights that merely investing in SCMITI is not enough and managers need to ensure that IT investments are leveraged to create unique capabilities that enhance resilience, rather than simply acquiring technology. It paves the way for managers to choose appropriate IT-based platforms and tools for enhancing SCR, by suggesting those that enhance IT-driven SCCs such as flexibility, visibility, responsiveness, collaboration and integration across the supply chain.

Moreover, the results indicate that to enhance SCR, business managers need to invest not only in suitable technologies that directly contribute to IT-driven SCCs but also in the security of these digital systems. Managers need to integrate robust risk management practices, including cybersecurity measures, into their IT strategy and ensure the continuation of the IT-driven SCCs that ultimately lead to SCR. With the increasing sophistication of cyber threats, managers should prioritise cybersecurity as an integral part of their SCRM strategy, especially when implementing and leveraging IT systems across the supply chain. This will help in effectively leveraging IT capabilities and enhancing resilience.

Also, managers should be aware that SCSC can have both positive and negative impacts on resilience. Given the interconnected nature of these findings, managers should adopt a holistic approach to SCM. This includes aligning IT investments with capability development, integrating risk management practices and considering the SCSC.

6.3 Limitations and future research

While we are confident in the validity and reliability of our study, some potential limitations should be acknowledged. The data were collected from supply chain professionals in the industry sector, which may limit the generalisability of the findings to the service sector. Moreover, the data collection was done during the COVID-19 pandemic, which may render its findings less applicable over time, despite their relevance to the constructs explored.

Future research can utilise more modern theories on technology integration to delve deeper into the complex interplay among SCSC, SCCs and SCR to provide additional insights into the underlying mechanism of achieving SCR. In light of the growing interest in supply chain digitalisation and the use of generative AI, as well as the consolidation of product design in digital artefacts, examining the integration of cybersecurity with SCR is a crucial research direction. Ghadge *et al.* (2019) provide multiple future research directions for cybersecurity and SCR, further highlighting the importance of this area. Another area worthy of investigation is the potential of digitalisation to enhance sustainability, despite the significant greenhouse gas emissions that may arise from cloud computing.

7. Conclusion

The complex global supply chain structure, combined with interconnected IT systems, has made businesses more vulnerable to various disruptions, such as geopolitical tensions and cyberattacks. This study is motivated by the urgent need to understand how SCMITI can enhance SCR, given the frequent disruptions and their significant economic impact on businesses and society worldwide. We utilised both DCV and RBV as the theoretical lens to conceptualise the interplay between SCMITI and SCR. We formulated IT-driven SCCs as mediators and SCSC and SCRM as two moderators. We developed six hypotheses and collected data through a survey from supply chain professionals. We analysed data and tested the hypotheses using PLS-SEM. The outcomes of this empirical research not only provide industry professionals with knowledge on converting SCMITI into SCR but also significantly enrich the academic literature on SCR. Our empirical analysis indicates that IT-driven SCCs serve as essential driving forces in fostering SCR, while SCRM functions as a crucial catalyst that reinforces this association.

Note

1 BRICS is the acronym representing the emerging national economies of Brazil, Russia, India, China and South Africa.

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Appendix

Table A1 Summary analysis of the measurement model: Cronbach's alpha, composite reliability and average variance extracted

Latent construct: SCMITI					
Labels	*DC* ($\alpha = 0.81$, CR = 0.86, AVE = 0.51)	Literature	Loading	*t*-value	Weight
DC 1	Our company uses automatic data capture systems (e.g. bar code) across the supply chain	Rai *et al.* (2006)	0.72	21.09	0.24
DC 2	We have common definitions of key data elements (e.g. customer, order, part number) across the supply chain		0.76	26.33	0.23
DC 3	Our company stores the same data (e.g. order status) in different databases across the supply chain consistently		0.63	13.08	0.20
DC 4	Our company uses an application portfolio and services (e.g. ERP, ASP, reusable software modules/components, emerging technologies, etc.)	Lu and Ramamurthy (2011)	0.74	24.71	0.24
DC 5	Our company uses data management services and architectures (e.g. databases, data warehousing, data availability, storage, accessibility, sharing, etc.)		0.73	20.52	0.24
n.a.	*Our company uses radio frequency identification (RFID) technology across the supply chain*	Guo *et al.* (2014)	*n.a.*	*n.a.*	*n.a.*
DC 6	Our company applies international data content standards (e.g. ISO 9000) to have strong data consistency in the supply chain	van den Hoven (2004)	0.70	22.72	0.25
Labels	*CFAI* ($\alpha = 0.78$, CR = 0.85, AVE = 0.53)	Literature	Loading	*t*-value	Weight
CFAI 1	Our company uses supply chain planning applications (e.g. demand planning, transportation planning and manufacturing planning)	Rai *et al.* (2006)	0.79	37.93	0.30
CFAI 2	Our company uses supply chain transaction applications (e.g. order management, procurement, manufacturing and distribution)		0.79	26.81	0.27
CFAI 3	Our company uses supply chain applications with internal applications of our organization (such as enterprise resource planning (ERP))		0.66	13.72	0.21
CFAI 4	Our company uses customer relationship applications with internal applications of our organization		0.72	20.30	0.25
CFAI 5	Our company builds consistent interoperable, cross-functional department databases to enable concurrent engineering, rapid experimentation and simulation and co-creation	Yu *et al.* (2018)	0.68	18.92	0.33
Latent construct: SCCs					
Labels	*SCFL* ($\alpha = 0.69$, CR = 0.81, AVE = 0.52)	Literature	Loading	*t*-value	Weight
SCFL 1	Our company has flexibility in production in terms of the volume of the order and production schedule	Chowdhury and Quaddus (2017)	0.74	23.83	0.36
n.a.	*Our company produces different types of products to meet customer requirements*		*n.a.*	*n.a.*	*n.a.*
SCFL 2	Our company has the multi-skilled workforce to continue production		0.65	15.43	0.38
n.a.	*Our company has contract flexibility such as partial order, partial payment, partial shipment, etc.*		*n.a.*	*n.a.*	*n.a.*
SCFL 3	Our company has flexibility in sourcing		0.72	18.26	0.29
SCFL 4	Our company has flexibility in distribution		0.76	24.74	0.37
n.a.	*Our company is capable of introducing a new product*		*n.a.*	*n.a.*	*n.a.*
Labels	*SCVI* ($\alpha = 0.79$, CR = 0.86, AVE = 0.55)	Literature	Loading	*t*-value	Weight
SCVI 1	Our supply chain partners have the information for monitoring and changing operations strategy	Mandal *et al.* (2016)	0.73	22.92	0.26
SCVI 2	Our supply chain partners have access to inventory, and order status information for forecasting		0.75	21.52	0.23
SCVI 3	Our supply chain partners have the necessary information system for tracking goods		0.75	24.91	0.26
SCVI 4	Our company has a business intelligence system	Chowdhury *et al.* (2019)	0.73	21.71	0.29
SCVI 5	Our company tracks the information on operations in the supply chain		0.74	29.18	0.32

(***continued***)

Table A1

Labels	*SCRS* ($\alpha = 0.81$, CR = 0.88, AVE = 0.64)	Literature	Loading	*t*-value	Weight
n.a.	*Our company's supply chain is order-driven rather than forecast-driven*	Braunscheidel and Suresh (2009)	*n.a.*	*n.a.*	*n.a.*
SCRS 1	Our company's supply chain is capable of responding to real market demand		0.79	33.90	0.32
SCRS 2	Our company's supply chain is capable of forecasting market demand		0.84	42.20	0.32
SCRS 3	Our company's supply chain can leverage the competencies of our partners to respond to market demands		0.83	39.51	0.34
SCRS 4	Our company's supply chain can respond to changes in demand without overstock or lost sales		0.73	21.14	0.28
Labels	*SCCL* ($\alpha = 0.87$, CR = 0.91, AVE = 0.71)	Literature	Loading	*t*-value	Weight
SCCL 1	Our company works jointly with its key suppliers to achieve mutual goals	Mandal *et al.* (2016)	0.84	50.49	0.31
SCCL 2	Our company develops strategic objectives jointly with our supply chain partners		0.88	53.97	0.30
SCCL 3	Our company shares rewards and risks evenly with our supply chain partners		0.80	33.39	0.28
SCCL 4	Our company jointly works with its key supply chain members for mutual benefits		0.86	40.83	0.30
Labels	*SCIN* ($\alpha = 0.77$, CR = 0.85, AVE = 0.53)	Literature	Loading	*t*-value	Weight
SCIN 1	Our company has real-time integration and connection among all internal functions (i.e. from raw material management through production, shipping and sales)	Huo (2012)	0.77	28.68	0.29
SCIN 2	Our company has fast real-time searching of information (e.g. the level of inventory, logistics-related operating data)		0.74	22.20	0.30
SCIN 3	Our company shares our demand forecast with our major supplier		0.73	24.50	0.25
SCIN 4	Our company shares production and delivery schedules across the supply chain	Rai *et al.* (2006)	0.75	26.28	0.30
SCIN 5	Our downstream partners (e.g. distributors, wholesalers and retailers) share their actual sales data with us		0.63	14.07	0.24
Latent construct: SCR					
Labels	*SCAG* ($\alpha = 0.72$, CR = 0.82, AVE = 0.54)	Literature	Loading	*t*-value	Weight
n.a.	*Our company is fairly sensitive to the opportunities and threats in the business environment*	Liu *et al.* (2018)	*n.a.*	*n.a.*	*n.a.*
SCAG 1	Our company can rapidly respond to the changing market		0.78	25.10	0.38
SCAG 2	Our company reserves extra service capacity in response to the rapidly changing market		0.79	29.03	0.37
SCAG 3	Our company continuously tries to discover additional needs for our customers of which they are unaware		0.76	26.78	0.36
SCAG 4	Our company fully authorizes its managers to make special accommodations for important clients		0.59	10.84	0.22
Labels	*SCSU* ($\alpha = 0.87$, CR = 0.91, AVE = 0.61)	Literature	Loading	*t*-value	Weight
SCSU 1	Our company recovers resources in sustainable practices through various activities such as reuse, recycling, selling of scrap and waste material	Kusi-Sarpong *et al.* (2019)	0.71	20.44	0.19
SCSU 2	Our company invests a proportion of its total budget in doing research to support sustainable products production		0.75	22.66	0.21
SCSU 3	Our company has capabilities to package, label and transport products in an environmentally friendly manner		0.80	34.99	0.20
SCSU 4	Our company uses the adoption of innovative manufacturing practices to minimize energy consumption and waste in production		0.80	32.63	0.22

(***continued***)

Table A1

SCSU 5	Our company has responsiveness to customers and market demand and awareness of using environmentally friendly and green products benefits		0.84	43.77	0.23
SCSU 6	Our company implements and enforces the socio-environmental standards and policies		0.80	33.50	0.23
Labels	*SCRE* ($\alpha = 0.79$, CR = 0.86, AVE = 0.61)	Literature	Loading	*t*-value	Weight
SCRE 1	In the case of disruption, our company can recover in a short time	Chowdhury and Quaddus (2017)	0.82	37.10	0.38
SCRE 2	In the case of disruption, our company can absorb a huge loss		0.65	13.98	0.24
SCRE 3	In the case of disruption, our company can reduce the impact of loss by our ability to handle crisis		0.84	37.74	0.35
SCRE 4	In the case of disruption, our company can recover from the crisis at less cost		0.82	36.38	0.31
Labels	*SCRM* ($\alpha = 0.85$, CR = 0.90, AVE = 0.69)	Literature	Loading	*t*-value	Weight
SCRM 1	Our company has a department or a team to manage supply chain risks and disruptions	Ambulkar *et al.* (2015)	0.78	25.04	0.24
SCRM 2	Our company has KPIs and metrics to monitor supply chain risk		0.85	40.88	0.32
SCRM 3	Our company has an information system in place to manage supply chain risks and disruptions		0.87	54.79	0.33
SCRM 4	Our organization is detecting operation risks by monitoring the supplier, doing inspections and tracking	Sreedevi and Saranga (2017)	0.82	41.33	0.32
Labels	*SCSC* ($\alpha = 0.57$, CR = 0.77, AVE = 0.53)	Literature	Loading	*t*-value	Weight
SCSC 1	Our company tries to deal directly with buyers and suppliers to reduce the complexity of the supply chain	Chowdhury and Quaddus (2017)	0.75	19.40	0.52
n.a.	*Our company does not have much forward and backward flow of goods and services in our supply chain*		*n.a.*	*n.a.*	*n.a.*
SCSC 2	Our company uses multiple suppliers to avoid the risk of supply		0.69	12.59	0.36
SCSC 3	Our company has multiple buyers to avoid the buyers' disruptions		0.76	17.04	0.49

Source: Authors' own work

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